practical Work_2
Data Analysis and Visualization

Due Date: 04/03/2025

1. Data Collection.
   i. Using the last dataset scraped from the book2scrape (last lab dataset)
   ii. Your dataset must have at least 500 rows and multiple features (columns).

   Suggested Websites for Web Scraping:
   - https://books.toscrape.com/
2. Analyze the dataset:
   i. Generate a word cloud using the book title (check the highest keyword in the titles.)
   ii. Using any unsupervised learning algorithms of your choice (See lecture note 2: K-means, KNN, PCA, etc.) to learn patterns in the dataset.
   iii. Determine the best or optimal cluster for the dataset
   iv. Display the result for optimal cluster using Elbow methods; Silhouette and Davies-Bouldin index.
   v. Save the new dataset (including column: cluster, Genre and Genre label)
   vi. Plot the book cluster visualization for the genre
3. Classification task
   i. Use MLP classifier with varying hyperparameters such as: hidden_layer, epoch / max iterations, learning rate, activation functions, optimizer, etc.
   ii. Evaluate the performance of the MLP classifier using metrics like: Accuracy, recall, precision, f1-score.
   iii. Display the results using confusion matrix, and t-SNE.

For Submission:

- A Python notebook (.ipynb) or script (.py) containing code.
- A CSV file containing the scraped/ extracted dataset.
- A report (PDF) summarizing your results of your findings with screenshots of your results.